Algebraic geometry 2026

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Contents

Chapter 1

Mynotes from 2026 class

Some references used here:

- Mumford's red book [2]
- Humphreys Linear algebraic groups [1]

1.1 Lecture 1 and 2

First some definitions and lemmas worth to remember.

- (Integral extension). Let R be a ring and $S \subset R$ a subring. We say that R is integrally dependent over S if every $r \in R$ is integral over S, that is, there exist a monic polynomial $f \in S[X]$ such that $f(r) = 0$.

  Lemma 1.1.1. Let $S \subset R$ be rings. A finite subset of elements $b_1, \ldots, b_m \in R$ are all integral over S if and only if $S[b_1, \ldots, b_m]$ is finitely generated (as a $S-$module) over S.

  Proof: ($\Rightarrow$) Let $b \in R$ be integral over S, i.e., there exist

  $$f = X^d + a_1 X^{d-1} + \cdots + a_d \in S[X] \tag{1.1}$$

  such that $f(b) = 0$. Then, using long division, we see that $S[b] = S + \cdots + Sb^{d-1}$ which is finitely generated. Now, if $b_1, \ldots, b_m$ are all integral over S, we can argue by induction that $S[b_1, \ldots, b_{m-1}]$ is finitely generated over S. However, since b_m is integral over $S[b_1, \ldots, b_{m-1}]$, $S[b_1, \ldots, b_m]$ is finitely generated over $[b_1, \ldots, b_{m-1}]$ and it must be finitely generated over S aswell.

  ($\Leftarrow$) Suppose that $S[b_1, \ldots, b_m] = S\beta_1 + \cdots + S\beta_n$ for some $\beta_i \in R$. Then for any $b \in R$ which fixes $S[b_1, \ldots, b_m]$, we have $b\beta_i = \sum_j a_{ij}\beta_j$ for some $(a_{ij}) \in \mathrm{Mat}_n(S)$. Therefore

  $$0 = (bI - (a_{ij}))^*(bI - (a_{ij}))v = \det(bI - (a_{ij}))v \tag{1.2}$$

where $v = (\beta_1, \ldots, \beta_n)^T$. Therefore $\det(bI - (a_{ij}))\beta_i = 0, \forall i$ and since $1 \in S\beta_1 + \cdots + S\beta_n$, we see that $\det(bI - (a_{ij})) = 0$. Since $\det(X - (a_{ij})) \in S[X]$ is monic, we are done. □

Corollary 1.1.2. (Transitive property of integral dependence). Given rings $R \supset S \supset T$, if R is integral over S and S is integral over T, then R is integral over T.

Question: Do we need Noetherian hypothesis using this statement?

Proof: Let $r \in R$, then by hypothesis there exist $f = X^d + s_1 X^{d-1} + \cdots \in S[X]$ such that $f(r) = 0$. Define $A = T[s_1, \ldots, s_d]$, then $A[r]$ is a finitely generated $A-$module because $S \subset A$ and lemma (1.1.1). Now, since A is a finitely generated $T-$module by lemma (1.1.1), we see that $A[r]$ is a finitely generated $T-$module. Now, assuming T is Noetherian we see that $T[r]$ is finitely generated as a $T-$module. □

Remark 1.1.3. We can certainly say (using the previous proof) that if $r \in R$, then $S[r]$ is finitely generated as a $T-$module.

Lemma 1.1.4. Let $K \supset k$ (k a perfect field) be a finitely generated extension field with transcendence degree $\partial_k(K) = d$, then there exist generators $z_1, \ldots, z_d, z_{d+1}$ such that

(1) $K = k(z_1, \ldots, z_{d+1})$
(2) $\{z_1, \ldots, z_d\}$ are algebraicaly independent
(3) z_{d+1} is separable over $k(z_1, \ldots, z_d)$

Lemma 1.1.5. (Noether's normalization lemma - Mumford's red book). Let R be a finitely generated algebra over k. Assume also that R is an integral domain. If R has transcende degree n over k (which means that $\partial_k \text{Frac}(R) = n$), then there exist $x_1, \ldots, x_n \in R$ algebraicaly independent and such that R is integral over $k[x_1, \ldots, x_n]$.

TODO: Study prove this version of the lemma using the statement made in class.

Lemma 1.1.6. (Noether's normalization lemma - Statement made in class). Let R be a finitely generated $k-$algebra. Then there exist $x_1, \ldots, x_n \in R$ algebraically independent over k such that R is integral over $k[x_1, \ldots, x_n]$.

Proof (Nagata): Since R is finitely generated as a k-algebra, there exist generators $y_1, \ldots, y_m$ such that $R = k[y_1, \ldots, y_m]$. Now suppose that $y_1, \ldots, y_m$ are algebraically independent, then we can take $x_i = y_i$. If not, then there exist $p \in k[Y_1, \ldots, Y_m] \setminus 0$ such that $p(y_1, \ldots, y_m) = 0$. Write

$$p(Y_1, \ldots, Y_m) = \sum_{\alpha \in \mathbb{Z}^m_{\geq} 0} a_\alpha Y_1^{\alpha_1} \cdots Y_m^{\alpha_m} \tag{1.3}$$

and define a new polynomial

$$\begin{aligned} q(Y_1, \ldots, Y_m) &= p(Y_1, Y_2 + Y_1^{r_2}, \ldots, Y_m + Y_1^{r_m}) \\ &= \sum_\alpha a_\alpha (Y_1^{\alpha_1 + r_2\alpha_2 + \cdots + r_m\alpha_m} + \cdots) \end{aligned} \tag{1.4}$$

for some choice of positive integers $r_2, \ldots, r_m$. Take a positive number $b > \alpha_i$, $\forall \alpha_i$ such that $a_\alpha \neq 0$. If we take $r_i = b^i$, then $\alpha_1 + \alpha_2 b + \cdots + \alpha_m b^{m-1}$ are all different numbers in base b. So, from different powers of the form $\alpha_1 + r_2\alpha_2 + \cdots + r_m\alpha_m$ we have a maximal one $d = \max\{\alpha_1 + \cdots r_m\alpha_m : a_\alpha \neq 0\}$ and

$$q(Y_1, \ldots, Y_m) = aY_1^d + \text{smaller degree terms} \in k[Y_2, \ldots, Y_m][Y_1], \ a \neq 0. \tag{1.5}$$

Now, if we let $z_i = y_i - y_1^{r_i} \in R, i = 2, \ldots, m$, we see that $q(y_1, z_2, \ldots, z_m) = p(y_1, \ldots, y_m) = 0$ and y_1 is integral over $k[z_2, \ldots, z_m]$. By induction on number of generators, there exist $x_1, \ldots, x_n$ algebraically independent such that $k[z_2, \ldots, z_m]$ is integral over $k[x_1, \ldots, x_n]$. So, by lemma 1.1.1 and its corollary (1.1.1), we see that $k[y_1, z_2, \ldots, z_m]$ is integral over $k[x_1, \ldots, x_m]$. Now, by definition, it's clear that $k[y_1, z_2, \ldots, z_m]$ is a finitely generated $R-$module. So by the corollary (1.1.2) again, we see that R is integral over $k[x_1, \ldots, x_m]$. $\square$

Lemma 1.1.7. Let $S \subset R$ be a ring extension. If R is a field, which integral over S, then S is a field.

Proof: Let $s \in S \setminus 0$ and $r \in R$ be its inverse. Since R is integral over S, there exist $f = X^d + a_1 X^{d-1} + \cdots \in S[X]$ be such that $f(r) = 0$. Therefore

$$0 = s^{d-1} f(r) = r + a_1 + sa_2 + \cdots + s^{d-1}a_0 \Rightarrow r \in S. \tag{1.6}$$

$\square$

Remark 1.1.8. Let $L|K$ be a field extension. Then

$$\begin{tikzcd} L|K \text{ finite extension} \arrow[r, bend left] & L|K \text{ algebraic extension} \arrow[l, bend left, "+ \text{ finitely generated extension}"] \end{tikzcd}$$

Theorem 1.1.9. (Zariski). Let k be a field and $\mathfrak{m} \subset R = k[X_1, \ldots, X_m]$ be a maximal ideal. Then $L = R/\mathfrak{m}$ is a finite extension field of k.

Proof: By Noether normalization lemma (1.1.6), there exist $a_1, \ldots, a_n \in L$ algebraically independent such that L is integral over $k[a_1, \ldots, a_n]$. By lemma (1.1.7), we conclude that $k[a_1, \ldots, a_n]$ is a field. Suppose that $n > 0$, so a_1 has an inverse. This means that $a_1 g(a_1, \ldots, a_n) = 1$ for some $g \in k[X_1, \ldots, X_n]$, which constradicts the fact that $a_1, \ldots, a_n$ are alegrbaically independent. Then we conclude that L is an integral (and then algebraic) over k. However, since L is a finitely generated extension ($L = k(\overline{X}_1, \ldots, \overline{X}_m)$) over k we conclude that L is in fact a finite field extension over k. $\square$

Theorem 1.1.10. Suppose that k is an algebraically closed field. Let $I \subset R = k[X_1, \ldots, X_m]$ be an ideal, then $V(I) = \varnothing \iff I = k[X_1, \ldots, X_m]$.

Proof: If $I \neq R$, then there exist a maximal ideal $\mathfrak{m} \supset I$. So $V(I) \supset V(\mathfrak{m})$ and it's sufficient to prove that $V(\mathfrak{m}) \neq \varnothing$. By theorem (1.1.9), we have that $L = R/\mathfrak{m}$ is a finite $k-$extension, so we must have $L = k$ because k is algebraically closed. So there exist $a_i \in k, i = 1, \ldots, m$ such that $X_i - a_i \in \mathfrak{m}$ and $(X_1 - a_1, \ldots, X_m - a_m) \subset \mathfrak{m}$. However $\varphi : R \to k : F \mapsto F(a_1, \ldots, a_m)$ has kernel $\mathrm{Ker}(\varphi) = (X_1 - a_1, \ldots, X_m - a_m)$ and since k is a field, $R/\mathrm{Ker}(\varphi)$ must be a field and $\mathrm{Ker}(\varphi)$ is a maximal ideal. Therefore we conclude that $\mathfrak{m} = (X_1 - a_1, \ldots, X_m - a_m)$. Finally $V(X_1 - a_1, \ldots, X_m - a_m) = \{a_1, \ldots, a_m\} \neq \varnothing$. $\square$

Remark 1.1.11. The hypothesis $k = \overline{k}$ in the previous theorem is crucial. For example if $k = \mathbb{R}$, we have $V(x^2 + 1) = \varnothing$.

Corollary 1.1.12. From the proof of the previous theorem, we see that there exist a bijection

$$\begin{aligned} k^n &\longleftrightarrow \begin{Bmatrix} \text{maximal ideals} \\ \mathfrak{m} \subset R \end{Bmatrix} \\ (a_1, \ldots, a_n) &\longmapsto (X_1 - a_1, \ldots, X_n - a_n) \end{aligned} \tag{1.7}$$

Theorem 1.1.13. (Hilbert Nullstelensatz). Let $A \subset k[X_1, \ldots, X_n]$ be an ideal, then

$$IV(A) = \sqrt{A}. \tag{1.8}$$

Proof: The inclusion $\sqrt{A} \subset IV(A)$ is easy. Now, assume $g \in IV(A)$, where $A = (f_1, \ldots, f_m)$. This means that we have the hypothesis:

$$f_i(a_1, \ldots, a_n) = 0,\ \forall i = 1, \ldots, m \Rightarrow g(a_1, \ldots, a_n) = 0\ (*) \tag{1.9}$$

Now, define the ideal

$$I = Ak[X_1, \ldots, X_{n+1}] + (1 - g(X_1, \ldots, X_n)X_{n+1}). \tag{1.10}$$

If I is a proper ideal, then by corollary (1.1.12), we have $I \subset (X_1 - a_1, \ldots, X_{n+1} - a_{n+1})$, which is the kernel of the evaluation map $k[X_1, \ldots, X_{n+1}] \to k : f \mapsto f(a_1, \ldots, a_{n+1})$. This means that

$$\begin{cases} f_i(a_1, \ldots, a_n) = 0,\ \forall i = 1, \ldots, m \\ g(a_1, \ldots, a_n)a_{n+1} = 1 \end{cases} \tag{1.11}$$

which contradicts our initial hypothesis $(*)$. Therefore, I is not a proper ideal, so $1 \in I$ and there exists $h_1, \ldots, h_{m+1} \in k[X_1, \ldots, X_{n+1}]$ such that

$$1 = \sum_{j=1}^{m} h_j(X_1, \ldots, X_{n+1}) f_j(X_1, \ldots, X_n) + h_{m+1}(X_1, \ldots, X_{n+1})(1 - g(X_1, \ldots, X_n)X_{n+1}). \tag{1.12}$$

Now, by projecting this equation onto $k[X_1, \ldots, X_{n+1}]/(1 - gX_{n+1})$, we get

$$g^{\ell}(X_1, \ldots, X_n) = \sum_{j=1}^{m} h_j(X_1, \ldots, X_{n+1}) g^{\ell} \times f_j(X_1, \ldots, X_n) \tag{1.13}$$

modulo $(1 - gX_{n+1})$, where $H_j = h_j(X_1, \ldots, X_{n+1})g^{\ell} \in k[X_1, \ldots, X_n]$, $\forall j$ if $\ell \gg 0$. Since $\{\overline{X}_1, \ldots, \overline{X}_{n+1}\}$ is algebraically free in $k[X_1, \ldots, X_{n+1}]/(1 - gX_{n+1})$, we conclude that, in fact, we have equality

$$g^{\ell}(X_1, \ldots, X_n) = \sum_{j=1}^{m} H_j(X_1, \ldots, X_n) \times f_j(X_1, \ldots, X_n) \tag{1.14}$$

in $k[X_1, \ldots, X_n]$. □

Remark 1.1.14. We could have used the homomorphism

$$k[X_1, \ldots, X_{n+1}] \to k(X_1, \ldots, X_n) : X_{n+1} \mapsto g,\ X_i \mapsto X_i,\ i \neq n+1 \tag{1.15}$$

at the end of the previous proof.

1.2 Lecture 3

Definition 1.2.1. We define an affine algebraic set $V \subset k^n$ to be a set of the form $V = V(S)$ for some $S \subset k[\underline{X}]$.

Notation 1.2.2. Let V be an affine algebraic set. We write

$$D_V(g) = \{p \in V : g(p) \neq 0\} = V \setminus V(g). \tag{1.16}$$

These open sets are called "principal open subsets".

Lemma 1.2.3. Any open subset $U \subset V$ of an affine algebraic set is quasicompact.

Proof: Consider a covering $U = \bigcup_{\lambda \in \Lambda} U_\lambda$. Observe that $V \setminus U = V(g_1, \ldots, g_m)$, so we have the equality $U = \bigcup_{i=1}^{m} D_V(g_i)$. In particular, we see that principal open subsets form a basis for the Zariski topology on V. Therefore, it's sufficient to prove that a principal open subset is quasicompact. So we assume $U = D_V(g)$. Furthermore, we have $U_\lambda = \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i})$ and

$$U = \bigcup_{\lambda\in\Lambda} \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i}). \tag{1.17}$$

So, without lost of generalizaiton, we can assume $U = \bigcup_{\lambda\in\Lambda} D_V(g_\lambda)$ from beginning. Now

$$\begin{aligned} D_V(g) = \bigcup_{\lambda\in\Lambda} D_V(g_\lambda) \iff \\ V\cap V(g) = V\cap V(\langle g_\lambda : \lambda\in\Lambda\rangle) \iff \\ V\cap V(g) = V\cap V(\langle g_\lambda : \lambda\in I\rangle)\ (I\subset\Lambda \text{ finite}). \end{aligned} \tag{1.18}$$

In the last equation we used Hilbert basis theorem: The sequence

$$\langle g_{\lambda_1}\rangle \subset \langle g_{\lambda_1}, g_{\lambda_2}\rangle \subset \cdots \tag{1.19}$$

must terminate at some point, because $k[\underline{X}]$ is Noetherian. Then we conclude that $D_V(g) = \bigcup_{\lambda\in I} D_V(g_\lambda)$. □

Definition 1.2.4. (Regular functions). Let $V \subset k^n$ be an affine algebraic set and $U \subset V$ be an open set. A map $\varphi : U \to k$ is said to be *regular* if for any $p \in U$, there exist $p \in U' \subset U$, polynomials $f, g \in k[\underline{X}]$ such that $g(q) \neq 0$, $\forall q \in U'$ and $\varphi|_{U'} = f/g|_{U'}$.

Lemma 1.2.5. Let $V \subset k^n$ be an affine algebraic set and $g \in k[X_1, \dots, X_n]$. Then the ser of regular functions $\varphi : D_V(g) \to k$ is of the form f/g^m for some $f \in k[X_1, \dots, X_n]$ and $m > 0$. So

$$\mathcal{F}(D_V(g)) \cong k[X_1, \dots, X_n]_g \tag{1.20}$$

where $\mathcal{F}$ denotes the sheaf of regular functions of V.

Proof: First we have $D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i)$ and since φ is regular, there exists $f_i, h_i \in k[\underline{X}]$, $i = 1, \dots, \ell$ such that

$$q \in D_V(g_i) \Rightarrow h_i(q) \neq 0 \text{ (so } q \in D_V(h_i)),\ \varphi\big|_{D_V(g_i)} = \frac{f_i}{h_i}\bigg|_{D_V(g_i)}. \tag{1.21}$$

Now, we have two fundamental relations:

$$\begin{cases} D_V(g_i) \subset D_V(h_i),\ \forall i = 1, \dots, \ell \\ D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i) \end{cases} \tag{1.22}$$

Using $V = V(F_1, \dots, F_m)$ and Hilbert Nullstelensatz (1.1.13), we get

$$\begin{cases} \sqrt{(F_1, \dots, F_m, g_i)} \subset \sqrt{(F_1, \dots, F_m, h_i)}, \ \forall i = 1, \dots, \ell \quad (*) \\ \sqrt{(F_1, \dots, F_m, g)} = \sqrt{(F_1, \dots, F_m, g_1, \dots, g_\ell)} \quad (**) \end{cases} \tag{1.23}$$

From $(*)$, we deduce that $g_i^{n_i} = \sum_j A_j F_j + B_i h_i$ and restricting to V we get $g_i^{n_i} = B_i h_i$. Now, since $\varphi|_{D_V(g_i)} = \frac{B_i f_i}{g_i^{n_i}}$ and $D_V(g_i) = D_V(g_i^{n_i})$, we can change notation $B_i f_i \to f_i$ and $g_i^{n_i} \to g_i$.

Now we do a trick: since $g_j f_i - f_j g_i = 0$ on $D_V(g_i) \cap D_V(g_j) = D_V(g_i g_j)$, we get

$$g_i g_j (g_j f_i - f_j g_i) = 0 \ \text{ on } V. \tag{1.24}$$

Furthermore, we still have $\varphi|_{D_V(g_i)} = g_i f_i / g_i^2$. So we change notation again: $g_i^2 \to g_i$ and $g_i f_i \to f_i$, so we have

$$g_i f_j - g_j f_i = 0, \ \forall i, j = 1, \dots, \ell \quad (***). \tag{1.25}$$

Now, by equation $(**)$, there exist $m > 0$ such that

$$g^m = \sum_j A_j F_j + \sum_k B_k g_k \tag{1.26}$$

and we get on V:

$$g^m = \sum_k B_k g_k. \tag{1.27}$$

Now, by using equation $(***)$, we have

$$f_i g^m = \sum_k B_k f_i g_k = \sum_k B_k f_k g_i = (\sum_k B_k f_k) g_i. \tag{1.28}$$

So if $q \in D_V(g_i)$, we have $\varphi(q) = \frac{f_i}{g_i}(q) = \frac{f}{g^m}(q)$, where we set $f = \sum_k B_k f_k$. $\square$

Chapter 2

Some notes from Humphreys

2.1 Pre - varieties and varieties

2.1.1 Projective varieties

Definition 2.1.1. Let V be a vector space, we define its associated projective variety by

$$\mathbb{P}(V) = (V \setminus 0)/\sim \tag{2.1}$$

where we let $v \sim w \iff v = \lambda w$ for some $\lambda \in k^{\times}$. Suppose that $\dim(V) = n+1$, then by choosing a basis of V, we identify $V = k^{n+1}$ and we write $\mathbb{P}(V) = \mathbb{P}^n$.

We define the Zariski topology on $\mathbb{P}$ by letting a closed set to be a locus of a homogeneous polynomial. So $X \subset \mathbb{P}^n$ is closed if and only if $X = V(S)$ for some collection $S \subset k[X_0, \dots, X_n]^h$ of homogeneous polynomials. We have a set of isomorphisms ($i = 0, \dots, n$):

$$\phi_i : k[X_1, \dots, X_n] \to k[X_0, \dots, X_n]^h : g \mapsto X_i^{\deg(g)} g(\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i} \dots, \frac{X_n}{X_i}). \tag{2.2}$$

The inverse of this map takes $\phi_i : h(X_0, \dots, X_n) \mapsto h|_{X_i=1}$. In fact if $h = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^{n+1}} a_\alpha X^\alpha \in k[X_0, \dots, X_n]$ is homogeneous of degree d, then

$$\begin{aligned}
\psi_i \phi_i(h) &= X_i^d h(\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, 1, \frac{X_{i+1}}{X_i}, \dots, \frac{X_n}{X_i}) \\
&= \sum_\alpha a_\alpha X_1^{\alpha_1} \cdots X_{i-1}^{\alpha_{i-1}} X_i^{d - \sum_{j \neq i} \alpha_j} X_{i+1}^{\alpha_{i+1}} \cdots X_n^{\alpha_n} \\
&= h, \text{ (because } \sum_j \alpha_j = d).
\end{aligned} \tag{2.3}$$

Conversely

$$\phi_i\psi_i(g) = X_i^d g(\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i})\Big|_{X_i=1} = g. \tag{2.4}$$

A homogeneous coordinate of $\mathbb{P}^n$ is a map

$$f_i : U_i = \{[x_0, \dots, x_n] : x_i \neq 0\} \to k^n : [x_0, \dots, x_n] \mapsto (\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}). \tag{2.5}$$

Since $U_i = \mathbb{P}^n \setminus V(x_i)$, it's an open set by definition. Furthermore, the map f_i is an isomorphism. In fact, let $Y \subset U_i$ be a closed set, then $\overline{Y}$ is closed in $\mathbb{P}^n$, so there exist $S \subset k[X_0, \dots, X_n]^h$ such that $\overline{Y} = V(S)$. So we claim that $f_i(Y) = V(\psi_i(S))$. In fact, we have

$$\begin{aligned} x = (x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in V(\psi_i(S)) &\iff \phi_i(h)(x) = 0 \ \forall h \in S \\ &\iff h[x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n] = 0 \ \forall h \in S \\ &\iff h f_i^{-1}(x) = 0 \ \forall h \in S \\ &\iff f_i^{-1}(x) \in V(S) \cap U_i = Y \\ &\iff x \in f_i(Y). \end{aligned} \tag{2.6}$$

Conversely, let $Z = V(T) \subset k^n$. We want to show that $f_i^{-1}(Z) \subset U_i$ is closed. We claim that $f_i^{-1}(Z) = V(\phi_i(T)) \cap U_i$, which can be proved in the sames lines as before.

Lemma 2.1.2. (Affine characterization for closedness). A subset $X \subset \mathbb{P}^n$ is closed if and only if $X \cap U_i$ is closed in U_i for all $i = 0, \dots, n$.

Lemma 2.1.3. Consider the map

$$\begin{aligned} \varphi : &\mathbb{P}^m \times \mathbb{P}^n \longrightarrow \mathbb{P}^{(m+1)(n+1)-1} \\ &([x_0, \dots, x_m], [y_0, \dots, y_n]) \longmapsto [\dots, x_i y_j, \dots] \end{aligned} \tag{2.7}$$

where we use $[\dots, x_i y_j, \dots]$ to abreviate $[x_0y_0, \dots, x_0y_n, \dots, x_my_0, \dots, x_my_n]$. We claim that $\varphi(\mathbb{P}^m \times \mathbb{P}^n) \subset \mathbb{P}^{(m+1)(n+1)-1}$ is a closed embedding, so it's an isomorphism onto its image.

Proof: Use U_i, V_j, W_{kl} to denote the affine open charts of $\mathbb{P}^m, \mathbb{P}^n$ and $\mathbb{P}^{(m+1)(n+1)-1}$ respectively. The strategy is to show that $\varphi(U_i \times V_j)$ is closed in W_{ij}. However, if $(x, y) \in U_i \times V_j$, then $\varphi(x, y) = [\dots, x_k y_l, \dots]$, where we assume $x_i = 1, y_j = 1$. Now, we observe that the image is of the form $[\dots, u_{kl}, \dots]$, where $u_{kl} = x_k y_l$. Then we conclude that $\varphi(U_i \times V_j)$ is the locus

$$f_{ij}^{-1}(\{[u_{kl} : (k,l) \neq (i,j), \ u_{kl} - u_{kj}u_{il} = 0, \ \forall k, l]\}) \overset{\text{closed}}{\subset} W_{ij}. \tag{2.8}$$

It remains to show that φ is injective. To do this, it's sufficient to see that $\varphi|_{U_i \times V_j}$ (viewed as a map to $\varphi(U_i \times V_j)$) can be inverted. The equation is just

$$[\dots, u_{kl}, \dots] \mapsto ([u_{0j}, \dots, u_{mj}], [u_{i0}, \dots, u_{in}]). \tag{2.9}$$

$\square$

Corollary 2.1.4. If $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$ are closed subsets, then $\varphi(X \times Y) \subset \mathbb{P}^{(m+1)(n+1)-1}$ is a closed subset.

2.1.2 Varieties

Chapter 3

Algebraic Groups

3.1 Basic definitions

Some important observations:

- If $\mathbb{F}$ is a finite field, the map which takes a polynomial in $\mathbb{F}[X_1, \ldots, X_n]$ and send to a function $\mathbb{F}^n \to \mathbb{F}$ is not an isomorphism. For example, if $\mathbb{F}$ has q elements, then $\prod_{q\in\mathbb{F}}(X-q) \in \mathbb{F}[X] \mapsto 0$. However, if $\mathbb{F}$ is infinity and $a_nX^n+a_{n-1}X^{n-1}+\cdots+a_0 = 0$, then we can select distinc elements $\alpha_0, \ldots, \alpha_n$ and we conclude that

$$\begin{pmatrix} a_0 & \ldots & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \alpha_0 & \alpha_1 & \cdots & \alpha_n \\ \vdots & \vdots & \cdots & \vdots \\ \alpha_0^n & \alpha_1^n & \cdots & \alpha_n^n \end{pmatrix} = 0. \tag{3.1}$$

  However $\det(\alpha_i^j)_{i,j=0}^n = \prod_{i<j}(\alpha_i - \alpha_j) \neq 0$ and we conclude that $a_0, \ldots, a_n = 0$. Now, if $f \in \mathbb{F}[X_1, \ldots, X_n]$, then we can write (assumin X_1 appears in f):

$$f = g_d(X_2, \ldots, X_n)X_1^d + \cdots + g_1(X_2, \ldots, X_n)X_1 + g_0(X_2, \ldots, X_n). \tag{3.2}$$

  So the equality $f = 0$, the previous step for the case $\mathbb{F}[X]$ and an induction on the number of variables yields $g_d, \ldots, g_0 = 0$. Then we see that if $|\mathbb{F}| = \infty$, we can identify the ring $\mathbb{F}[X_1, \ldots, X_n]$ with polynomial functions $\mathbb{F}^n \to \mathbb{F}$.

  As a consequence of this story, we see that if $|\mathbb{F}| = \infty$, then

$$I(\mathbb{F}^n) = (0) \subset \mathbb{F}[X_1, \ldots, X_n]. \tag{3.3}$$

- (Hilbert basis theorem). The Hilbert basis theorem states that if A is noetherian, then $A[X]$ is noetherian. Since $\mathbb{F}[X]$ is a PID (and in consequence noetherian), we conclude by an induction on the number of variables that $\mathbb{F}[X_1, \ldots, X_n]$ is noetherian.

Remark 3.1.1. A ring is said to be noetherian if it satisfy the ascending chain condition, or equivalently, if all of its ideals are finitely generated.

- (Algebraic sets). Let $S \subset \mathbb{F}[X_1, \ldots, X_n]$, then $V(S) = \{a \in \mathbb{F}^n : f(a) = 0\}$. It's immediate that $V(S) = V((S))$. The relation $V(\bigcup_{i \in I} S_i) = \bigcap_{i \in I} V(S_i)$ allow us to define the Zariski topology on $\mathbb{F}^n$. Moreover, for any subset $V \subset \mathbb{F}^n$, we let $I(V) = \{f \in \mathbb{F}[X_1, \ldots, X_n] : f(a) = 0,\ \forall a \in V\}$. Then we have

$$\begin{cases} VI(W) = \overline{W} \\ IV(\mathfrak{a}) = \sqrt{\mathfrak{a}} \end{cases} \tag{3.4}$$

We also have the important relation

$$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}). \tag{3.5}$$

In fact

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \underset{\mathfrak{a}\cap\mathfrak{b} \subset \mathfrak{a},\mathfrak{b}}{\subset} V(\mathfrak{a} \cap \mathfrak{b}) \underset{\mathfrak{a}\mathfrak{b} \subset \mathfrak{a}\cap\mathfrak{b}}{\subset} V(\mathfrak{a}\mathfrak{b}) = V(\{fg : f \in \mathfrak{a}, g \in \mathfrak{b}\}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b}). \tag{3.6}$$

- (Coordinate rings). Let $\mathfrak{a} = I(V)$ where V is an algebraic set. We define the coordinate ring $A[V] = \mathbb{F}[X_1, \ldots, X_n]/\mathfrak{a}$.

- (LEX order). We define an ordering on $\mathbb{Z}^n_{\geq 0}$ by $\alpha > \beta$ if for the smallest i such that $\alpha_i \neq \beta_i$ we have $\alpha_i > \beta_i$. So, with a fixed LEX on $\mathbb{F}[X_1, \ldots, X_n]$, we can definte the leading term of a polynomial: $\mathrm{LT}(f) = a_\alpha X^\alpha$, where α is the smallest tuple, relative to LEX such that $a_\alpha \neq 0$. In the case $\mathrm{LT}(f) = a_\alpha X^\alpha$, we also write $\mathrm{LM}(f) = X^\alpha$.

- (Groebner Basis). Let $\mathfrak{a}$ be an ideal in $\mathbb{F}[X_1, \ldots, X_n]$. A groebner basis of $\mathfrak{a}$ is a finite subset $G \subset \mathfrak{a}$ such that $(\mathrm{LT}(g) : g \in G) = \mathrm{LT}(\mathfrak{a})$.

  Example 3.1.2. Let $R = \mathbb{F}[X_1, \ldots, X_n, Y_1, \ldots, Y_n]$ with LEX: $X_1 > \cdots > X_n > Y_1 > \cdots > Y_n$ and define an ideal $\mathfrak{a} = (X_1 - g_1, \ldots, X_n - g_n)$ for $g_i \in \mathbb{F}[Y_1, \ldots, Y_n]$. Since $\mathrm{LT}(X_i - g_i) = X_i$, it's clear that $(X_1, \ldots, X_n) \subset \mathrm{LT}(I)$. Now, if $f \in R$ and $X_j \nmid \mathrm{LT}(f)$, $\forall j$, then $f \in \mathbb{F}[Y_1, \ldots, Y_n]$ because its smallest term has power $\alpha \in \{0\} \times \mathbb{Z}^n$. So, we need to show that $I \cap \mathbb{F}[Y_1, \ldots, Y_n] = (0)$. This can be seen by proving that the kernel of $\varphi : R \to \mathbb{F}[Y] : X_i \mapsto g_i, Y_i \mapsto Y_i$ is I. The only nontrivial inclusion is $\mathrm{Ker}(\varphi) \subset I$. To see this use a division algorithm to write

$$f = h_1(X_1 - g_1) + \cdots + h_n(X_n - g_n) + r \tag{3.7}$$

  where $r \neq 0$ implies that no term of r is divisible by $\mathrm{LT}(X_i - g_i) = X_i$, $\forall i$. By applying φ on both sides of this equation we have the desired inclusion. From this we see that $G = (X_1 - g_1, \ldots, X_n - g_n)$ is a groebner basis for I.

- (Hilbert Polynomial).

Definition 3.1.3. Let $I \subset \mathbb{F}[X_1, \ldots, X_n]$ be an ideal. We deffine the affine Hilbert function:

$$^a\mathrm{HF}_I : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0} : s \mapsto \dim_{\mathbb{F}}(\frac{\mathbb{F}[X_1, \ldots, X_n]_{\leq s}}{I_{\leq s}}). \tag{3.8}$$

Example 3.1.4. For example, the number $^a\mathrm{HF}_{(0)}(s)$ is the number of n-tuples of nonnegative integers whose sum is $\leq s$. To compute this, write the generating function

$$\prod_{i=1}^{n} \frac{1}{1 - X_i} = \sum_{\alpha \in \mathbb{Z}^n_{\geq 0}} X^\alpha. \tag{3.9}$$

Now put $X_i = t$, $\forall i$. We see that

$$(1-t)^{-n} = \sum_{s \geq 0} \binom{-n}{s} (-1)^s t^s = \sum_{s \geq 0} \binom{n+s}{s} t^s. \tag{3.10}$$

So $^a\mathrm{HF}_{(0)}(s) = \binom{n+s}{s} = \frac{1}{n!} s^n + \cdots$.

Example 3.1.5. Let $f \in \mathbb{F}[X_1, \ldots, X_n]$ be a degree d polynomial. Then

$$^a\mathrm{HF}_{(f)}(s) = \binom{s+n}{s} - \binom{s-d+n}{s-d} = \frac{d}{(n-1)!} s^{n-1} + \cdots. \tag{3.11}$$

An important property of the affine Hilbert function is the following:

Theorem 3.1.6. Let $I \subset \mathbb{F}[X_1, \ldots, X_n]$ be an ideal, then

$$^a\mathrm{HF}_I(s) = {}^a\mathrm{HF}_{\mathrm{LT}(I)}(s), \ s \geq 0. \tag{3.12}$$

Another main result is the existence of the Hilbert polynomial.

Theorem 3.1.7. There exist a polynomial $^a\mathrm{HP}_I(t) \in \mathbb{Q}[t]$ such that

$$^a\mathrm{HP}_I(s) = {}^a\mathrm{HF}_I(s), \ \forall s \geq 0. \tag{3.13}$$

Proof (sketch):

- Find a Groebner basis G of I
- Let $M \subset \mathbb{Z}_{\geq 0^n}$ be such that $\{X^\beta\}_{\beta \in M} = \{\mathrm{LM}(g) : g \in G\}$
- Write
$$C(I) = \bigcap_{\beta \in M} C(\beta), C(\beta) = \{\alpha | X^\beta \nmid X^\alpha\} \tag{3.14}$$
as a union of $C(J, \tau) = \{\alpha | \alpha_j = \tau(j) \ \forall j \in J\}$. To do this use
$$C(\beta) = \bigcup_{i=1}^{n} \bigcup_{t_i=0}^{\beta_i - 1} C(\{i\}, \ \tau : i \mapsto t_i). \tag{3.15}$$

– Use the formula $|C(J,\tau)_{\leq s}| = \binom{d+s-|\tau|}{d}, d = n - |J|, |\tau| = \sum_{j\in J}\tau(j)$ and the inclusion-exclusion principle to compute

$$^a\mathrm{HP}_I(s) = |C(I)_{\leq s}| = \left|\bigcup_{(J,\tau)} C(J,\tau)_{\leq s}\right|. \tag{3.16}$$

Example 3.1.8. Let $I = (X_1 - X_3^2, X_2 - X_3^3)$ and fix the LEX order with $X_1 > X_2 > X_3$. Then we calculate

– $G = \{X_1 - X_3^2, X_2 - X_3^3\}$

– $M = \{(1,0,0),(0,1,0)\}$

– We have

$$\begin{aligned} C(I) &= C(\{1\},\tau : 1 \mapsto 0) \cap C(\{2\},\tau : 2 \mapsto 0) \\ &= C(\{1,2\},\tau : 1 \mapsto 0, 2 \mapsto 0). \end{aligned} \tag{3.17}$$

So we conclude that $|C(I)|_{\leq s} = \binom{1+s-0}{1} = s+1$.

- (Algebraic dimension). Here we define the dimension of an algebraic set.

Definition 3.1.9. Let $V \subset \mathbb{F}^n$ be an algebraic set. We define its dimension as $\dim(V) = \deg {}^a\mathrm{HP}_{I(V)}(t)$. It has the property that it's the biggest integer $d \geq 1$ such that there exist indexes $1 \leq i_1 < \cdots < i_d \leq n$ such that $I(V) \cap \mathbb{F}[X_{i_1},\ldots,X_{i_d}] = (0)$.

Example 3.1.10. Let $H_f = V(f)$, f irreducible, then

$$\begin{aligned} \dim(H_f) &= \deg {}^a\mathrm{HP}_{I(H_f)}(t) \\ &= \deg {}^a\mathrm{HP}_{(f)}(t) \\ &= \deg(\frac{\deg(f)}{(n-1)!}t^{n-1} + \cdots) \\ &= n-1. \end{aligned} \tag{3.18}$$

The dimension has the expected property: If $W \subset V$ is closed, then $\dim(W) \leq \dim(V)$. In fact

$$\begin{aligned} &I(V) \subset I(W) \Rightarrow \\ &I(V)_{\leq s} \subset I(W)_{\leq s} \Rightarrow \\ &{}^a\mathrm{HF}_{I(W)}(s) \leq {}^a\mathrm{HF}_{I(V)}(s),\ \forall s \geq 0 \Rightarrow \\ &\deg {}^a\mathrm{HF}_{I(W)}(t) \leq \deg {}^a\mathrm{HF}_{I(V)}(t). \end{aligned} \tag{3.19}$$

- (Transcendence degree). Let A be a field which is also a finitely generated $\mathbb{F}$-algebra. We define the transcendence degree of A over $\mathbb{F}$ (and write $\partial_{\mathbb{F}}(A)$) to be largest size of an algebraicaly independent subset of A over $\mathbb{F}$.

Example 3.1.11. We have $\partial_{\mathbb{F}}(\mathbb{F}[X_1, \ldots, X_n]) = n$. Indeed, let $\{f_1, \ldots, f_r\}$ be polynomials and suppose that $r > n$. Write

$$\begin{aligned} &W_s = \{f^\beta : \beta \in \mathbb{Z}^r_{\geq 0},\ |\beta| \leq s\},\ d = \max\{\deg(f_i) : i = 1, \ldots, r\} \\ &V_s = \{X^\alpha : |\alpha| \leq s\} \end{aligned} \tag{3.20}$$

Then $W_s \subset V_{ds}$ and

$$|W_s| = \binom{r+s}{s} \overset{s \gg 0}{>} \binom{n+sd}{sd} = \dim(V_{sd}) \geq \dim(W_s). \tag{3.21}$$

So, for $s \gg 0$ there exist $\{a_\beta\}_\beta \subset \mathbb{F}$ such that $\sum_\beta a_\beta f^\beta = 0$ and $\{f_1, \ldots, f_r\}$ is algebraicaly dependent.

Theorem 3.1.12. Let V be an algebraic set and write $I = I(V)$, then

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \partial_{\mathbb{F}}(\frac{\mathbb{F}[X_1, \ldots, X_n]}{I}). \tag{3.22}$$

Proof: Let $d = \dim(V)$, so there exist indexes $1 \leq i_1 < \cdots < i_d \leq n$ such that $I \cap \mathbb{F}[X_{i_1}, \ldots, X_{i_d}] = (0)$. This is equivalent to say that $[X_{i_1}], \ldots, [X_{i_d}] \in A[V]$ are algebraicaly independent over $\mathbb{F}$. Therefore $d \leq \partial_{\mathbb{F}}(A[V])$. Now, suppose that $\varphi_i = [f_i] \in A[V], i = 1, \ldots m$ are algebraicaly independent over $\mathbb{F}$ and put $N = \max \deg(f_i)$, then the map

$$\mathbb{F}[Y_1, \ldots, Y_m]_{\leq s} \mapsto \frac{\mathbb{F}[X_1, \ldots, X_n]_{\leq Ns}}{I_{\leq Ns}} : g \mapsto g(\varphi_1, \ldots, \varphi_m) \tag{3.23}$$

is injective. Then, by taking dimensions we get

$$\binom{m+s}{s} \leq {}^a\mathrm{HF}_I(s),\ \forall s \geq 0. \tag{3.24}$$

We know that both expressions are polynomials in s, so by taking degrees we conclude that $m \leq \dim(V) = d$ and $\partial_{\mathbb{F}}(A[V]) \leq d$. $\square$

With small changes in the above proof, we can verify that in the case of $A[V]$ being an integral domain, we have also $\partial_{\mathbb{F}}(K) = \dim(V),\ K = \mathrm{Frac}(A[V])$.

Example 3.1.13. Let $C = V(X_1 - X_3^2, X_1 - X_3^3)$, then $I(C) = (X_1 - X_3^2, X_2 - X_3^3)$ and

$$\frac{\mathbb{F}[X_1, X_2, X_3]}{I(C)} \cong \mathbb{F}[X_3] \tag{3.25}$$

so $\partial_{\mathbb{F}}(A[C]) = 1$.

Example 3.1.14. The purpose of this example is to show that if $V \subset \mathbb{F}^n, |\mathbb{F}| = \infty$ is a linear space, then $\dim(V) = \dim_{\mathbb{F}}(V)$.

– Let $\mathrm{Ann}(V) = \mathrm{Span}(f_1, \ldots, f_m)$. Since V has finite dimension, we have $v \in V \iff f(v) = 0\ \forall f \in \mathrm{Ann}(V)$ and $V = V(\{f_1, \ldots, f_m\})$.

– Let $f_i = \sum a_{ij} X_j$, where we identify X_j with the jth coordinate functions in $\mathbb{F}^n$. Now, if $A = (a_{ij})$ and apply reduction to put A in the form:

$$\begin{pmatrix} a_{1j_1} & * & 0 & * & \cdots & 0 & \cdots \\ & & a_{2j_2} & * & \cdots & 0 & \cdots \\ & & & & \vdots & \cdots & \vdots \\ & & & 0 & \cdots & a_{mj_m} & \cdots \end{pmatrix} \tag{3.26}$$

with $a_{ij_i} = 1$, $j_1 < j_2 < \cdots$. Now put LEX with: $X_{j_1} > \cdots X_{j_m} > X_j\ j \notin \{j_1, \ldots, j_m\}$ and write $f_i = X_{j_i} + h_i$ with $h_i \in \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}]$. Then we see that $A[V] \cong \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}]$ and $(f_1, \ldots, f_m)$ is prime. Therefore $I(V) = (f_1, \ldots, f_m)$. Finally

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \dim_{\mathbb{F}} \mathbb{F}[X_j : j \notin \{j_1, \ldots, j_m\}] = \dim_{\mathbb{F}} V. \tag{3.27}$$

3.2 Tangent Spaces

- (Regular maps). Let $V \subset \mathbb{F}^m, W \subset \mathbb{F}^n$ be algebraic sets. Then a map $f : V \to W$ is said to be regular if there exist polynomials such that $f = (f_1, \ldots, f_m)$. Such a map is continuous in the Zariski topology.

  Remark 3.2.1. We can identify $A[V]$ with the set of regular functions $V \to \mathbb{F}$. In this context, we have a contravariant functor:

  $$\varphi : V \to W \rightsquigarrow \varphi^* : A[W] \to A[V] : [g] \mapsto [g] \circ \varphi \tag{3.28}$$

  Example 3.2.2. Let $\varphi : \mathbb{F} \to C : x \mapsto (x^3, x^2, x)$ and write $C = (X_1 - X_3^2, X_2 - X_3^3)$. Remember the isomorphism $A[C] \to \mathbb{F}[Y] : [X_1] \mapsto Y^2, [X_2] \mapsto Y^3, [X_3] \mapsto Y$. So we have

  $$\varphi^* : A[C] \to A[\mathbb{F}] : [X_3] \mapsto [X_3] \circ \varphi. \tag{3.29}$$

  The first map is $[X_3] : C \to \mathbb{F} : (x^3, x^2, x) \mapsto x$ and the last map is $\mathbb{F} \to C : x \mapsto x$. At the algebraic level, the map φ^* is just the the isomorphism $\mathbb{F}[Y] \cong \mathbb{F}[X]$.

- (Tangent spaces). Let V be an algebraic set and $V_p = \mathbb{F}$ be the $A[V]-$module with action $[f]\alpha \equiv f(p)\alpha$. Let $I(V) = (f_1, \ldots, f_m)$. We define

  $$\begin{aligned} T_p(V) &= \{v \in \mathbb{F}^n : d(f)_p(v) = 0\ \forall f \in I(V)\} \\ &= \{v \in \mathbb{F}^n : d(f_i)_p(v) = 0\ \forall i\} \\ &= V(\{d(f_i)_p : i = 1, \ldots, m\}) \end{aligned} \tag{3.30}$$

Remark 3.2.3. Caution ! The polynomials of f_i here are not the generators of V, but of $I(V)$. For example, if $V = V(f)$, we can conlude that $T_p(V) = V(df_p)$ only if f is irreducible, i.e., if $I(V) = (f)$.

Here $df_p \equiv \sum_{i=1}^n D_i(f)(p)X_i \in \mathbb{F}[X_1, \dots, X_n]$. We have an isomorphism

$$T_p(V) \to \mathrm{Der}_{\mathbb{F}}(A[V], V_p) : v \mapsto D_v;\ D_v([f]) = \sum_{i=1}^n [D_i(f)] \cdot v_i \in \mathbb{F}. \tag{3.31}$$

with inverse $D \mapsto (D([X_i]))_{i=1}^n$. Of course $D_v([f]) = df_p(v)$.

Maybe it will be better to change notation of V_p to ${}^p\mathbb{F}$ or something.

Example 3.2.4. Let $C = (f_1, f_2),\ f_1 = X_1 - X_3^2, f_2 = X_2 - X_3^3$ and $p = (x^3, x^2, x) \in C$. We have

$$d(f_1)_p = 1 - 2xX_3,\ d(f_2) = 1 - 3x^2X_3 \tag{3.32}$$

and $T_p(C) = \mathbb{F}\langle(3x^2, 2x, 1)\rangle$.

Example 3.2.5. The determinant is an irreducible polynomial. Indeed, let $\det \in \mathbb{F}[X_{ij} : 1 \leq i, j \leq n]$. Then, if we apply Laplace theorem on the first row, we get

$$\det = A_{11}X_{11} - A_{12}X_{12} + \cdots + (-1)^n A_{1n}X_n \tag{3.33}$$

where A_{ij} denotes the determinant of the matrix obtained by excluding the ith row and jth collumn. By induction on the number of variables, we assume that A_{ij} are irreducible polynomials. Now, suppose that $\det = fg$, where $\deg(f), \deg(g) > 0$. Choose LEX with $X_{ij} > X_{k\ell}$ if $i < k$ or if $i = k$ and $j < \ell$. Then

$$\mathrm{LT}(A_{11})X_{11} = \mathrm{LT}(\det) = \mathrm{LT}(f)\mathrm{LT}(g). \tag{3.34}$$

Assume that $X_{11}|\mathrm{LT}(f)$ (so $\mathrm{LT}(g) < X_{11}$) and write $f = AX_{11} + B,\ \mathrm{LT}(B) < X_{11}$. Then we must have $AgX_{11} = A_{11}X_{11}$ and $A_{11} = Ag$. Since $\deg(g) > 1$ and A_{11} is irreducible, we must have $g = A_{11}$ and $A = 1$. Now we get

$$-A_{12}X_{12} + \cdots + (-1)^n A_{1n}X_n = BA_{11} \tag{3.35}$$

and $-\mathrm{LT}(A_{12})X_{12} = \mathrm{LT}(B)\mathrm{LT}(A_{11})$. Since $X_{22}|\mathrm{LT}(A_{11})$, we are led to $X_{22}|\mathrm{LT}(A_{12})$ which is a contradiction because X_{22} doenst appear in A_{12} by definition.

Now, since det is an irreducible homogeneous polynomial, the difference $\det -1$ must be irreducible. Therefore $I(SL_n\mathbb{F}) = (\det -1)$ and we can compute the tangent space

$T_I(SL_n) = V(d_I(\det))$. We have

$$\begin{aligned} d_I \det &= \sum_{k,\ell} D_{k\ell}(\det)(I) X_{k\ell} \\ &= \sum_{k,\ell} \sum_{\sigma\in S_n} D_{k\ell} \prod_{i=1}^{n} X_{i\sigma(i)}(I) X_{k\ell} \\ &= \sum_{k,\ell} \sum_{\sigma\in S_n} \delta_{\ell,\sigma(k)} \prod_{i\neq k} \delta_{i,\sigma(i)} X_{k\ell} \\ &= \sum_{k,\ell} \delta_{k\ell} X_{k\ell} \quad (\delta_{\ell,\sigma(k)} \prod_{i\neq k} \delta_{i,\sigma(i)} \neq 0 \text{ only if } \sigma = 1) \\ &= \sum_{k} X_{kk}. \end{aligned} \tag{3.36}$$

Therefore $T_I SL_n(\mathbb{F}) = \mathfrak{sl}_n \mathbb{F}$, the Lie algebra of traceless matrices.

- (Gauss lemma). We state the Gauss lemma.

Lemma 3.2.6. Let R be a UFD ring (think in $\mathbb{F}[X]$), then $R[X]$ is a UFD.

Proof: Let f and $K = \mathrm{Frac}(R)$. Since $K[X]$ is a UFD, we have $f = FG,\ F, G \in K[X]$ with F irreducible and $\deg(G) < \deg(f)$. Since R is a UFD, there exist $r, s \in R$ such that $rF, sG \in R[X]$. Decompose $rs = t_1 \cdots t_k$ into irreducible elements. Then if $F' = rF, G' = sG$, we have

$$[F'][G'] = [rs][f] = 0 \in R/(t_i)[X] \tag{3.37}$$

because $f \in R[X]$. So, since $R/(t_i)[X]$ is a domain, we see that $t_i | F'$ or $t_i | G'$. Then we see that there exist a partition $\{1, \dots, k\} = I \cup J$ such that $F'/\prod_{i\in I} t_i, G'/\prod_{j\in J} t_j \in R[X]$. It's clear that F' still irreducible, then an induction on degree yields the result. $\square$

Corollary 3.2.7. Let $f, g \in R = \mathbb{F}[X_1, \dots, X_n]$ and assume f irreducible with $f \nmid g$, then there exist $F, G \in R$ such that

$$Gf + Fg = h \in \mathbb{F}[X_1, \dots, X_{n-1}] \setminus 0. \tag{3.38}$$

Proof: Let $K = \mathrm{Frac}(\mathbb{F}[X_1, \dots, X_{n-1}])$, then $f, g \in K[X_n]$ and there exist $A, B \in K[X_n]$ such that $Af + Bg = 1$ (by Gauss lemma f still irreducible as an element of $K[X_n]$). Then there exist $h \in \mathbb{F}[X_1, \dots, X_{n-1}]$ such that $hA, hB \in \mathbb{F}[X_1, \dots, X_{n-1}]$. So by writing $G = hA, F = hB$ we are done. $\square$

Notation 3.2.8. Let $A = \mathbb{F}[X_1, \dots, X_n]/I$, where M is a $A-$module. We write

$$\mathcal{T}_{A,M} = \{(v_1, \dots, v_n) \in M^n : \sum_i [D_i(f)] \cdot v_i,\ \forall f \in I\}. \tag{3.39}$$

We have the isomorphism $\mathcal{T}_{A,M} \to \mathrm{Der}_{\mathbb{F}}(A, M) : v \mapsto D_v$ with inverse

$$D \mapsto (D(\overline{X_i}))_{i=1}^{n}. \tag{3.40}$$

Now consider the following context, let $K = \mathrm{Frac}(A)$ as A-module (A acts by multiplication). Then we have an isomorphism

$$\mathrm{Der}_{\mathbb{F}}(K, K) \to \mathrm{Der}_{\mathbb{F}}(A, K) : D \mapsto D|_A. \tag{3.41}$$

In particular we also have the isomorphism $\mathcal{T}_{A,K} \cong \mathrm{Der}_{\mathbb{F}}(K, K)$. We observe that this is also a $K-$vector space isomorphism, it's a bijection compatible with the $K-$action. From this isomorphism, we see that $\mathcal{T}_{A,K}$ only depends on K, and the fact that $\mathrm{Frac}(A) = K$.

Theorem 3.2.9. Let $A = \mathbb{F}[X_1, \ldots, X_n]/I$ be a domain and $K = \mathrm{Frac}(A)$. Then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathrm{Der}_{\mathbb{F}}(K, K). \tag{3.42}$$

Proof: Let $d = \partial_{\mathbb{F}}(A) = \partial_{\mathbb{F}}(K)$, then there exist $z_1, \ldots, z_{d+1}$ such that

- $K = \mathbb{F}(z_1, \ldots, z_{d+1})$
- $\{z_1, \ldots, z_d\}$ is algebraicaly independent
- z_d is separable over $\mathbb{F}(z_1, \ldots, z_d)$

Let $R = \mathbb{F}[z_1, \ldots, z_{d+1}] \subset K$ and $f \neq 0$ irreducible such that $f(z_1, \ldots, z_{d+1}) = 0$. Then by corollary (3.2.7), we see that $\alpha : K[Y_1, \ldots, Y_{d+1}] \to K : g \mapsto g(z_1, \ldots, z_{d+1})$ has kernel (f). Therefore

$$\begin{aligned} \mathcal{T}_{R,K} &= \{(v_1, \ldots, v_{d+1}) \in K^{d+1} : \sum_i [D_i(f)] \cdot v_i = 0\} \\ &= \{(v_1, \ldots, v_{d+1}) \in K^{d+1} : \sum_i D_i(f)(z_1, \ldots, z_{d+1}) \cdot v_i = 0\}. \end{aligned} \tag{3.43}$$

We see that $D_i(f)(z_1, \ldots, z_{d+1}) = 0 \Rightarrow D_i(f) \in (f)$ and $D_i(f) = 0$. Since $f \neq 0$ and $|\mathbb{F}| = \infty$, we must have $D_i(f) \neq 0$ for some i, so $D_i(f)(z_1, \ldots, z_{d+1}) \neq 0$ for some i. Then we conclude that

$$\dim_K \mathrm{Der}_{\mathbb{F}}(K, K) = \dim_K \mathcal{T}_{R,K} = d = \partial_{\mathbb{F}}(K) = \partial_{\mathbb{F}}(A). \tag{3.44}$$

$\square$

Corollary 3.2.10. If $A = \mathbb{F}[X_1, \ldots, X_n]/(f_1, \ldots, f_m)$, then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathcal{T}_{A,K} = n - \mathrm{Rank}_K([D_i(f_j)]). \tag{3.45}$$

Example 3.2.11. Let $C = V(f_1, f_2), f_1 = X_1 - X_3^2, f_2 = X_2 - X_3^3$. We have

$$A[C] = \frac{\mathbb{F}[X_1, X_2, X_3]}{(f_1, f_2)} \to \mathbb{F}[X_3] : [g] \mapsto g(X_3^3, X_3^2, X_3). \tag{3.46}$$

So $A[C]$ is a domain and we can define $K = \mathrm{Frac}(A[C])$. We compute

$$\mathrm{Rank}_K([D_i(f_j)]) = \mathrm{Rank}_K \begin{pmatrix} 1 & 0 & -2[X_3] \\ 0 & 1 & -3[X_3] \end{pmatrix} = 2. \tag{3.47}$$

Then we conclude $\dim(V) = 3 - 2 = 1$.

- (differential of a regular map). Here we define the notion of differential of a regular map.

Definition 3.2.12. Let $\varphi : V \to W$ be a regular map between algebraic varieties. Also, let $p \in V, q = \varphi(p) \in W$. We define:

$$d\varphi_p : \mathrm{Der}_{\mathbb{F}}(A[V], V_p) \to \mathrm{Der}_{\mathbb{F}}(A[W], W_q) : D_v \mapsto D_v \circ \varphi^* \tag{3.48}$$

At the level of tangent spaces, we have:

$$\begin{tikzcd}
\mathrm{Der}_{\mathbb{F}}(A[V], V_p) \arrow[rr, "d\varphi_p"] & & \mathrm{Der}_{\mathbb{F}}(A[W], W_q) \arrow[dd] & \\
& D_v \arrow[rr, mapsto] & & D_v \circ \varphi^* \arrow[dd, mapsto] \\
T_p(V) \arrow[uu] \arrow[rr, dashed] & & T_q(W) & \\
& v \arrow[uu, mapsto] \arrow[rr, mapsto] & & (D_v \circ \varphi^*([Y_i]))_{i=1}^m
\end{tikzcd}$$

In more details, if we write $\varphi = (f_1, \dots, f_m)$, then $\varphi^*([Y_i]) = [f_i]$ and we get

$$D_v([f_i]) = \sum_{j=1}^m [D_j(f_i)] \cdot v_j = \sum_{j=1}^m D_j(f_i)(p) v_j. \tag{3.49}$$

Therefore

$$d\varphi_p(v) = (D_j(f_i)(p))_{i,j} \cdot v = J_p(\varphi) \cdot v. \tag{3.50}$$

Example 3.2.13. Let $G \subset \mathbb{F}^{n^2}$ be an algebraic group and $\mu : G \times G \to G$ be the product of matrices. We have $G \times G \subset \mathbb{F}^{n^2} \times \mathbb{F}^{n^2} \cong \mathbb{F}^{2n^2}$. Now, identify the first n^2

coordinate functions with X_{ij} and the second n^2 coordinate functions with Y_{ij}. So we have

$$\mu_{ij} = \sum_{\ell} X_{i\ell} Y_{\ell j}. \tag{3.51}$$

Then we have

$$\frac{\partial \mu_{ij}}{\partial X_{rs}}(I) = \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) = \delta_{ir}\delta_{sj}. \tag{3.52}$$

Therefore

$$d_{(I,I)}\mu(A,B) = (\sum_{r,s} \frac{\partial \mu_{ij}}{\partial X_{rs}}(I) A_{rs} + \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) B_{rs})_{ij} = (A_{ij} + B_{ij}) = A + B. \tag{3.53}$$

TODO:

1. Prove that $T_1 G$ is a Lie algebra. Remember the star operator.
2. Compute the Lie algebra of SL_n using the fact that $\det -1$ is an irreducible polynomial.

Bibliography

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